

FLEA NEWS 32

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Please find herewith:

---lists of additional literature for 1975 (list 19), 1976 (list 19), 1977 (list 17), 1978 (list 18), 1979 (list 18), 1980 (list 14), 1981 (list 12), 1982 (list 9), 1983 (list 7), 1984 (list 5), 1985 (list 3), 1986 (list 1)
---additions and changes to the mailing list

* * *

HARRY HOOGSTRAAL

24-II-1917 24-II-1986

We have been informed by Dr. Robert Traub that Dr. Harry Hoogstraal died of lung cancer in Cairo, Egypt, on February 24, his 69th birthday. Although he was best known for his exhaustive studies on ticks and arboviruses, entomologists, parasitologists and medical scientists were among the beneficiaries of many of his studies.

Dr. Hoogstraal was born in Chicago, Illinois. He received the Bachelor of Arts degree from the University of Illinois, Urbana, in 1938 and the Master of Science degree in 1942. During this time he organized and led four zoological expeditions to Mexico. In 1943, he joined the United States Army as Malariologist and Head of the Entomology Department, 4th Area Medical Laboratory, Northington General Hospital, and 19th Medical General Hospital, retiring in 1946 with the rank of Captain MSC. Later that year he headed the Philippine Zoological Expedition, a joint venture between the Philippine Government and the Field Museum of Natural History, Chicago, Illinois. Upon his return, he was appointed Medical Zoologist with the United States Navy Bureau of Medicine and Surgery and participated in the University of California Expedition to Africa and Madagascar in 1948-49. In 1949-50 he headed the Sudan (Equatoria) Sub-unit of the United States Naval Medical Research Unit No. 3 (NAMRU-3) and was appointed Head of Medical Zoology of NAMRU-3 in 1950, a position he held until his death. In 1959 he received the PhD from the London School of Hygiene and Tropical Medicine, the ScD from the same institution in 1971, and the Doctorate (honoris causa) from the Faculty of Medicine, Ain Shams University, Cairo, Egypt, in 1978.

From the beginning of his professional career, Dr. Hoogstraal was interested in the distribution of animals and disease. At NAMRU-3 he supervised a

program which concentrated on the interrelationships between vertebrate hosts, their ectoparasites and pathogens, especially viral and rickettsial, causing human and animal diseases. Parts of this program extended to all parts of the world, but most of his early studies were performed in Egypt and adjacent countries in the eastern Mediterranean area. Early on, it became apparent that knowledge of the fauna of Egypt was poor so he set out to conduct a natural history survey of the country that was almost unparalleled in the history of the natural sciences. As a result, Egyptian birds and mammals are now as well known as those of western Europe or the United States. Tick speciation, biology, ecology, and their relationship to disease were subjects of particular interest to him. In the early 1950's tick species were unidentifiable in the larval and nymphal stages and there was much nomenclatural confusion among the named adults. Working from the principal that each species has unique biological properties which determine the significance and nature of the role of the species in disease transmission, he initiated a program of defining the tick species in Egypt and the Sudan. To do so properly, the investigation extended into the neighboring countries and eventually to all of Africa, Europe, Asia, Australia, numerous oceanic islands, and frequently North and South America. More than 350 of the approximately 800 known species were studied, named and described, if new to science, or redescribed, if poorly known. Concomitant to these studies, the largest and most varied collection of ticks in the world was amassed and a rearing program was begun in order to obtain the immature stages that could be associated with known adults.

Although the bulk of the studies directed by Dr. Hoogstraal concerned themselves with ticks and their relationships to disease transmission, they produced enormous amounts of peripheral data and specimens which were gladly shared with specialists throughout the world. Literally hundreds of scientists were the beneficiaries of field studies conducted or supported by him through NAMRU-3. In addition, because of his extensive travels, he developed a keen appreciation of the zoogeographic relationships of not just ticks and their hosts but of animals in general which would strongly influence his later studies.

As a natural corollary of his studies, the Hoogstraal Library on Ticks and Tickborne Diseases was established. It is the largest library of its kind in the world and contains over 95,000 reprints and reports, as well as several thousand biomedical periodicals, textbooks and specialized scientific, zoological and geographical books. Since much work on ticks has been carried out in the U.S.S.R. and the eastern European countries, a translation section was organized and hundreds of translations have been produced, copies of which have been provided to the scientific community. Beginning in 1970, and extending through 1982, the library was catalogued in a 7-volume Bibliography of Ticks and Tickborne Diseases from Homer (about 800 B.C.) to 31 December 1981, which comes to 3576 pages. It includes all known publications on tickborne diseases, including diagnosis, treatment, epidemiology, etc., as well as tick taxonomy, systematics, morphology, biology, ecology, host relationships, behavior, distribution, physiology and biochemistry. Three thousand copies of each volume were distributed to libraries and scientists throughout the world.

Dr. Hoogstraal was a member, a life member, or a charter member of 32 professional societies. Over the years he served on a myriad of national and

international committees, panels, councils and delegations. He also served in a number of editorial posts and was an external examiner and visiting lecturer at universities in Egypt, India, Australia, South Africa, England and the United States. He was the recipient of a multitude of professional honors and delivered lectures on various topics relating to medical zoology and epidemiology to gatherings of scientists all over the world. Finally, in addition to the involvements listed above, he also authored or coauthored more than 500 scientific papers in various technical journals.

During his years in Egypt, Dr. Hoogstraal became a respected egyptologist, and his villa in Maadi, south of Cairo, abounded with artifacts acquired during his world travels. He was also a talented sculptor, and usually had a bust in progress when he could steal time from his travels and professional obligations.

Even with such awesome involvement in his profession, perhaps none of his contributions compares to the constant support and encouragement meted out to the hundreds of colleagues, young and old, with whom he corresponded. There is hardly an aspect of entomology, parasitology, or medical zoology today that remains untouched by his influence, directly or indirectly. That he was a respected professional is attested to by the fact that over 40 species of plants, reptiles, birds, mammals, insects, mites, ticks and protozoans bear the specific name hoogstraali in his honor.

In addition to being a scientist of the first order, Harry Hoogstraal was a naturalist who believed that his profession was an art as well as a science. This seems to be a view unshared by younger professionals, and perhaps his passing is symbolic of what is happening today in the natural sciences. His gentle voice and quiet dignity will be sorely missed by friends and colleagues alike.

---R.E.L.

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RECENTLY PUBLISHED

Coevolution of Parasitic Arthropods and Mammals. K. C. Kim, ed. 1985. (ISBN 0-471-08546-4) John Wiley & Sons, Inc., New York. 800 pp. \$56 (ESA members), \$69.95 (nonmembers).

Interactions between plants and animals have long been recognized by both botanists and zoologists, and the presence of terpenoids, alkaloids and various other compounds in some plants is thought to be an evolutionary response to phytophagous insects. Observations of these interactions over the years has gradually given rise to the theory of coevolution between plants and their herbivores involving such diverse entities as host specificity, pollination strategies and seed dispersal. Similar interactions between various groups of animals, also resulting in coevolutionary phenomena, are not easily demonstrated and appear in the earlier literature under the guise of host specificity. During the last 30 years there have been two symposia devoted to the host specificity of vertebrate parasites: the first at Neuchâtel, Switzerland, in 1957; the second in Paris, France, in 1981. In both instances the emphasis was directed toward the non-arthropod parasites of vertebrates.

The appearance of Adrian Marshall's The Ecology of Ectoparasitic Insects (1981) marked the first concerted effort to summarize the extensive literature on the subject of ectoparasites and their interrelationships with each other and with their hosts. While coevolution is not discussed as such, it is amply alluded to in various chapters under other headings, thus laying the groundwork for such a study as this. This book consists of 13 chapters, divided into 4 parts and concluded by 2 appendices and an index.

Part one contains 3 chapters that attempt to introduce the subject from different points of view. In chapter one Kim discusses the various aspects of the evolutionary relationships of parasitic arthropods and mammals, including host-parasite associations, parasitism as opposed to commensalism, and ectoparasites as opposed to endoparasites. The chapter continues with a consideration of the host as a habitat, the biology of parasitic arthropods and their population patterns and dynamics. Interactions between parasitic arthropods are discussed, as are interactions between them and their hosts. In conclusion, the evolutionary aspects of these associations are reviewed and some conclusions drawn concerning possible evolutionary pathways to the ectoparasitic condition.

Of the remaining 2 chapters, the first is a short consideration of coevolution as a process. It includes a table of general differences between plant and mammalian ectoparasites and discusses, in general terms, many of the considerations involved in the ectoparasite mode of existence. Chapter three reviews mammals as the evolutionary partners in coevolution, first by considering them as a habitat for ectoparasitic arthropods, and then by reviewing the origins and geological history of the class Mammalia. Approximately half of this chapter is devoted to a review of the diversity and geographical distribution of recent mammals.

Part two contains chapters 4 through 8, which deal with insect ectoparasites of mammals. In chapter four, which deals with patterns of insect parasitism in mammals, the authors point out that ectoparasitic insects of mammals constitute less than 0.5% of all insects and belong to only 7 orders. Of these the Anoplura, Mallophaga and Siphonaptera are exclusively ectoparasitic and are discussed in subsequent chapters. A small number of minor taxa belonging to the orders Dermaptera, Hemiptera, Coleoptera and Diptera are the subjects of this chapter. Adaptations and evolutionary strategies of these species are discussed, along with their host associations and geographical distributions. The chapter concludes with a consideration of parasitic adaptations, including convergence of body form and structure and a discussion of host associations and specificity.

Chapter five deals with the evolution and host associations in the order Anoplura. It begins with a review of adaptations and evolutionary trends imposed by the habitat provided by the host. This is followed by an analysis of the host-parasite associations, arranged by order of the host, and a discussion of host specificity. The chapter concludes with an analysis of the origin and evolution of the class Mammalia, the origin of host associations and their evolution, and a brief consideration of phylogenetic relationships and historical biogeography.

Chapter six is concerned with the evolution of the Mallophaga on mammals. Since most of the members of this order are associated with birds, the chapter

is shorter than most. Representatives of all three mallophagan suborders occur on mammals and the authors conclude that mammal-infesting mallophagans could be useful in tracing the evolutionary affinities of their hosts, as has been done with some groups of birds. They also postulate that less than half of the existing species of mammal-infesting mallophagans are currently known to science (thanks to the inherent resistance on the part of most mammalogists to collect ectoparasites).

Chapter seven is entitled Evolutionary Aspects of the Disjunct Distribution of Lice on Carnivora. After noting that endemic anoplurans are absent from terrestrial carnivores and endemic mallophagans are absent from marine carnivores, the author poses four questions, the answers to which may be reached by two alternative hypotheses. A detailed discussion of adaptive strategies and host associations of both groups on mammals in general is followed by a review of the classification and evolution of the order Carnivora, and a synthesis of louse associations and distribution within the order. In a section on colonization and extinction the author advances some interesting hypotheses to explain the current distribution of lice on carnivores, based on competition for available space on the host and the ability to adapt to the change to an aquatic habitat on the part of their arctoid (bear-like) hosts.

Chapter eight deals with the coevolution of fleas and mammals. After a brief commentary on the interactions between fleas and their mammalian hosts, the author addresses the matter of host specificity and distribution. This is done by dividing the species of fleas into a few categories that are determined by degrees of specificity. Thus, for example, ultraspecific fleas are those that are restricted to a single host species; generispecific fleas those that infest a genus, or group of closely related hosts, etc. There follows a short consideration of zoogeography and evolution within the order Siphonaptera and a presentation of evidence of convergence, divergence and various adaptations to hosts and habitats. In the discussion the author synthesizes information about the host relationships, zoogeography, evolution and coevolution, and morphological adaptations in fleas, and posits a series of conclusions which suggest that there is more host specificity in the Siphonaptera than has generally been supposed, and that much of what we know about the order indicates an intensely close association between fleas, their mammal hosts and coevolution.

Part three contains chapters 9 through 12 and deals with coevolution in the Acari. Chapter nine treats the evolution of mesostigmatid mites and begins with a description of the group and a short section on its preadaptation for parasitism. Since not all mesostigmatid mites are parasites, the author next discusses the diversity, distribution and phylogeny of the group and then progresses through its evolutionary strategies and adaptations for parasitism by dealing with the individual families and subfamilies. In conclusion, it is emphasized that mesostigmatid mites show an unusual ecological diversity while retaining a remarkably uniform habitus. The dermanissoid families in particular demonstrate a variety of ecological and vertebrate relationships which combine to make them the most successful mammal parasites among the Mesostigmata.

Chapter ten deals with tick and mammal coevolution with emphasis on the ixodid genus Haemaphysalis. The authors begin with the premise that ixodid ticks evolved as obligate parasites of reptiles in the late Paleozoic or early

Mesozoic eras and that their subsequent coevolution with birds and mammals has been conservative. The least conservative have been the species associated with livestock, man and dogs. The premammalian evolution of ticks is discussed, followed by sections on the evolution of mammalian ticks and host specificity. The bulk of this chapter deals with the coevolution of ticks and mammals using the genus Haemaphysalis as a model. After a brief discussion of generic characters, the species are segregated into categories, depending upon whether they are considered to be structurally primitive, intermediate or advanced, and the subgenera of each of these subdivisions are discussed. The summary includes a brief resumé of the distribution and host relationships for the Argasidae and the subfamilies of the Ixodidae.

Chapter eleven, entitled Prostigmata-Mammalia Validation of Coevolutionary Phylogenesis, begins with a section on coevolution and symbiosis in which it was deemed necessary to include a table of the definitions of such arcane terms as polyhospitality, polybioty, asymbiospecificity, symbiophylogeny, (A) symbio-specific and other semantic nonsense. These are then employed in subsequent sections to make a case for coevolution in 3 families of the Prostigmata. I leave it to other readers to determine the success of this author.

Chapter twelve is concerned with the evolution of astigmatid mites on mammals and begins with a brief consideration of the requirements of a parasitic way of life. It is pointed out that both constructive and regressive anatomical adaptations are required, in addition to various biological modifications, to be a successful parasite. The authors then proceed to review the more important families that are parasitic on mammals. They conclude that there is strong evidence for coevolution in this group, especially in parasites that spend their entire lives on the host.

Part four, entitled Overview, contains chapter thirteen, Parasitism and Coevolution: Epilogue, and 2 appendices. In chapter thirteen the author summarizes the contents of the preceding chapters and makes the case that coevolution, to varying degrees, does exist between parasitic arthropods and mammals. He also stresses that the process involves both interspecific interactions and strategies to promote coexistence with other associating species. The chapter concludes with a discussion of the origin and phylogeny of parasite-host associations and examples of resource and phylogenetic tracking models.

Appendix A is a list of parasitic arthropods associated with mammals and Appendix B is a list of mammals and their parasitic arthropods.

In spite of the somewhat restrictive title, this book contains a wealth of information of interest to entomologists and acarologists in general, and is a must in the library of medical and veterinary entomologists. It is well organized, printed in a pleasing typeface and is remarkably free of mechanical errors.

Although Professor Kim is listed as the editor, in fact he is the sole author of four of the chapters and a co-author of three more, in addition to his contributions as an editor. Both he and the other authors are to be congratulated on their treatment of this fascinating subject. ---R.E.L.

A Colour Atlas of Insect Tissues Via the Flea by Miriam Rothschild, Yosef Schlein and Susumo Ito, 1986. (ISBN 0 7234 0891 2) Wolfe Publishing Ltd., Wolfe House, 3 Conway Street, London W1P GHE, 184 pp. + 261 figs. (188 in colour). \$79.95

The title of this book quite accurately describes its content. After a brief preface and introduction, there are three pages entitled Techniques. Included here are explanations of the various techniques and equipment employed in obtaining the light microscope, stereoscan and transmission electron photomicrographs. There is also a glossary for electron micrographs, a description of fixing and staining methods, solutions and a short list of abbreviations. Following this are 14 detailed drawings, most originals by Y. Schlein, showing the principal internal organs of the flea, based upon both sexes of Xenopsylla cheopis. The remainder of the book is divided into 25 sections, each dealing with an organ system or other anatomical manifestation such as resilin, blood cells, the sensillum, endocrine glands, etc. Each section is prefaced by a brief introduction, followed by a series of illustrations. Where both black and white and coloured plates are used, the coloured plates come first. Structures in the coloured plates are numbered corresponding to a labeling key for each set of plates. Black and white plates are labeled with letters which are explained in the captions. Appendix I includes illustrations of structures not found in fleas such as flight muscles, compound eyes and wing scales. Appendix II illustrates various parasites and symbionts in the internal organs of fleas. These include helminths, protozoans and bacteria. Ten references to other works are listed and there is a comprehensive index.

Many of the illustrations are of sections of the European rabbit flea, Spilopsyllus cuniculi. However, in total, 20 species representing 7 families are shown in the plates.

The book is logically organized and the printer has done an excellent job with the colour plates. The authors are to be congratulated for the production of this remarkable volume and the contribution it makes to our understanding of flea anatomy. ---R.E.L.

Insects of Southern Africa. C. H. Scholtz and Erik Holm, eds. 1985. (ISBN 0409 10486 8 Limited edition, 0409 10487 6 Hard cover) Butterworths, Durban, 502 pp., price not available.

Although not of direct interest to the readers of Flea News, attention is called to the availability of this volume on the Southern Africa fauna. There has long been a need for such a work and, although it has been estimated that no more than 5% of the Southern African insect fauna is known to science, this work is a creditable treatment of the known fauna. In passing, it should be noted that the siphonapteran fauna of the region is quite well known thanks to the works of DeMeillon, Haeselbarth, Segerman, Davis, Hardy and Zumpt.

Fleas figure in this volume in two ways. First, the 5-page account of the order was written by Ms. Joyce Segerman of the South African Institute for Medical Research. Second, throughout the myriad of black and white figures a stylized silhouette of a cat flea is used to indicate scale. This system has been used previously in field guides to vertebrates, but to my knowledge this is a first in a book on insects. ---R.E.L.

The Oxford Dictionary of Natural History. Michael Allaby, ed. 1985. (ISBN 0-19-217720-6) Oxford University Press, Oxford, New York, 688 pp. \$38.00

Again, this book may be of only passing interest to many recipients of Flea News. However, as horizons in Biology and related disciplines expand, so, too, does the plethora of new terms and old terms with new definitions, creating the need for a new, updated lexicon on the average of every 5 to 10 years. The term "natural history" is not defined in this work, but Lincoln, Boxshall and Clark, 1982, A Dictionary of Ecology, Evolution and Systematics, define it as "The study of nature, natural objects and natural phenomena". It would seem that the editor had this definition in mind, since he allotted space for earth and atmospheric sciences, genetics, cell structure and function, biochemistry, parasitology and a few other disciplines. Fifty-six contributors and consultants participated in this compilation of over 12,000 entries.

It is axiomatic that no totally complete lexicon has ever been written, and likely never will be. The appearance of this volume does not alter the axiom, although the approach followed by the editor and contributors is somewhat different from most similar treatments. For example, the names of taxa in the higher classification of both plants and animals are included (but not the viruses, etc.). In some cases, even genera are discussed. However, the treatment is very uneven and neither the order nor any of the families of the Siphonaptera are mentioned, for example. Even the word "flea" only appears in association with "flea beetle" and "water flea", an insect and a crustacean, respectively. Still, the terminology is current, the coverage broad and the price reasonable, and it now resides with the eight other dictionaries and glossaries on my desk, and the 20 or 30 additional word books in my library.
---R.E.L.

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HIGHER CLASSIFICATION OF THE SIPHONAPTERA

It was Karl Jordan's view that the Hystrichopsyllidae were so distinctive that they formed a superfamily equal to "all other fleas put together". These were his actual words to me. Now K. J. had a sixth sense where systematics/morphology were concerned but he did not live long enough to learn that Hystrichopsylla and Stenoponia are set apart from all other fleas - as far as we know now - by the possession of polytrophic ovaries as opposed to panoistic ovaries. Is it not time for some enterprising flea specialist to section a few other genera and see how far this important character supports K. J.'s belief? In view of a current opinion that fleas are descended from a Mecopteran-like ancestor it is interesting that in that Order there are both families with polytrophic and panoistic ovaries.
---Miriam Rothschild

* * *

A PAIR OF CORRECTIONS: Your attention is called to p. 194 of Flea News 31 where Bob Elbel's article on flea larvae is listed as published in Bull. Soc. Vector Ecol. 19(1). The correct citation should be 10(1). The Bulletin also erred in omitting a portion of the abstract, and a corrigendum appears in Bull. Soc. Vector Ecol. 10(2): 126. Our apologies are added to those of the Bulletin and corrected citations appear in this issue.

A REQUEST - Dr. Jan Máca (Dept. Medical Entomology, Parasitology Institute, Czechoslovakian Academy of Science, Branišovská 31, 370 05 Č. Budějovice, Czechoslovakia) would greatly appreciate receiving preserved material - e.g., in spirit - of the larvae of various flea species (correctly determined., i.e., raised from eggs laid by determined adult fleas) for morphological comparison.

* * *

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ADDITIONS

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* * *

REPRINTS RECEIVED - We wish to thank the following for supplying reprints and citations since the publication of Flea News 31:

J. W. Amrine
J. C. Beaucournu
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O. Larson
P. M. Linardi
V. Mahnert
A. Olsen
W. Skuratowicz

* * *

Answer to FLEA TRIVIA QUESTION in Flea News 31: In the year 1950 87 new taxa were described!

A PREFATORY NOTE TO THE SPECIES LIST - In keeping with Mr. Smit's practice of numbering the taxa chronologically, the following list includes numbers for those species described in 1984. Readers are urged to call any omissions in these lists to our attention.

- 98401 biconcava Chen, Ji & Wu, 1984 Rhadinopsylla (Actenophthalmus)
98402 liupanshanensis Li, Wu & Liu, 1984 Nearctopsylla (Neochinopsylla)
98403 mytila Hürka, 1984 Lagaropsylla
98404 largificus Smit, 1984 Parapsyllus magellanicus
98405 euneomys Lewis, 1984 Agastopsylla nylota
98406 qinlingensis Zhang, Wu & Liu, 1984 Hystrihopsylla weida
98407 brevifrontata Zhang, Wu & Liu, 1984 Palaeopsylla
98408 anserocephoides Zhang, Wu & Liu, 1984 Palaeopsylla
98409 minutiprominula minutiprominula Zhang & Liu, 1984 Geusibia
98410 minutiprominula ningshanensis Zhang & Liu, 1984 Geusibia
98411 longihilla Zhang & Liu, 1984 Geusibia
98412 digitata Cai Wu & Liu, 1984 Callopsylla (Callopsylla)
98413 ventrisinulata Chen, Zhang & Liu, 1984 Leptopsylla (Pectinoctenus)
pectiniceps
98414 jinciensis Xiao, 1984 Ischnopsyllus (Hexactenopsylla)

PROVISIONAL LISTS OF TAXA DESCRIBED IN 1985 AND 1986

1985

- Peromyscopsylla himalaica australishaanxia Zhang & Liu
Peromyscopsylla scaliforma Zhang & Liu
Paradoxopsyllus calceiforma Zhang & Liu
Liuopsylla conica Zhang, Wu & Liu
Sigmactenus celebensis Lewis & Jones
Gryphopsylla hetera Lewis & Jones
Stivalius exoticus Lewis & Jones
Typhloceras poppei orientalis Mahnert & Beaucournu
Rhadinopsylla (Actenophthalmus) stenofronta Xie
Frontopsylla rotunditruncata Cai & Liu
Typhlomyopsyllus esinus Liu, Shi & Liu
Caenopsylla eremita Beaucournu & Kowalski
Dorcadia xijiensis Zhang & Dang
Macrostylophora furcata Shi, Liu & Wu
Ceratophyllus (Ceratophyllus) qinghaiensis Zhang & Ma
Vermipsylla parallela rhinopithecus Li
Frontopsylla (Orfrontia) postprojicia Liu, Qi & Li

1986

Coorilla allisoni Mardon

* * *

FLEA TRIVIA QUESTION: How many genera and species did Julius Wagner describe in his lifetime? Answer in Flea News 33!

(1975 - 1981)

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1975 (list 19)

Pizzimenti, J. J. - Evolution of the prairie dog genus *Cynomys*. Occas. Papers Mus. nat. Hist., Univ. Kansas 39: 1-73 [Siph.: 53-57]

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1976 (list 19)

*Hallas, T. E. - Lopper (Fleas). Natur og Museum 17(4): 1-22

McNeill, W. H. - Plagues and peoples. Anchor Press, New York, 340 pp.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1977 (list 17)

*Baker, K. P. - Hypersensitivity of the skin to fleas. Proc. 6th World Conf., World Small Animal Vet. Assoc.: 43-50

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1978 (list 18)

*Schumann, H. - Ordnung Siphonaptera - Flöhe. In: Urania Tierreich, Insekten, Leipzig, Jena, Berlin 4. Aufl., Urania Verlag: 409-415

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1979 (list 18)

Waage, J. K. - The evolution of insect/vertebrate associations. Biol. J. Linn. Soc. 12: 187-224

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1980 (list 14)

Bavaasan, A., S. Covelmaa, K. Chumarchan & S. Zigmed - Iz istorii izučeniya ektoparazitov životnykh - nositeley čumy v MNR. Probl. prir. očagovosti čumy, Irkutsk 1: 100-101

Covelmaa, C., Z. Bavaasan, S. Zigmed & K. Chumarchan - K parazitarnoj charakteristike pustynno-gobijskogo očaga čumy. Probl. prir. očagovosti čumy, Irkutsk 1: 67-68

Gund, N. - K izučeniju bloch melkich mlekopitajuščich Gobi-Altajskogo ajmaka MNR. Probl. prir. očagovosti čumy, Irkutsk 1: 97-99

Kozlovskaja, O. L. & S. A. Chamaganov - Ektoparazity melkich mlekopitajuščich okrain peskov Altan-Els i Borig-Del v MNR. Probl. prir. očagovosti čumy, Irkutsk 1: 94-95

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1981 (list 12)

*Dudich, A. - Blchy (Siphonaptera) drobných zemných cicavcov Zahorskej nížiny a Malých Karpát. Acta Rer. natur. Mus. nat. slov. Bratislava 28: 91-109

Haas, G. E. - Fleas (Siphonaptera: Pulicidae and Vermipsyllidae). In: Alaskan Wildlife Diseases. Dieterich, R. A., ed. Univ. Alaska, Fairbanks, Alaska 99701, p. 184-186

(1981 - 1983)

- Hackman, W. - Book review. Traub, R. & H. Starcke (ed.) 1980: Fleas. Proceedings of the International Conference on Fleas, Aston Wold, Petersborough, UK, 21-25 June 1977. - A. A. Balkema, Rotterdam, 1980, 420 s. Pris. Hfl. 95:-, ISBN 90-6191-018-8. Notulae Ent. 61:90

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1982 (list 9)

- Anonymous - Flea-bitten rat gives new clues to Black Death. New Scientist 94(1306): 492
- *Dudich, A. & A. Stollmann - Súčasný stav regionálneho faunistického výskumu drobných zemných cicavcov na Slovensku. Lynx 21: 67-78
- Girardi, C., O. Abate & A. Farca - L'impiego del fenthion per via diadermica nel trattamento delle infestazioni da ectoparassiti nel cane. Ann. Fac. Med. Vet. Torino 28: 272-278 [Published in 1984]
- Kamyszek, F. - The practical evaluation of insecticides and their application. Wiad. Parazytol. 28(1-2): 275-286
- *Kenyon, J. R. - The Willoughby Gardner Library. A collection of early printed books on natural history. National Museum of Wales, Cardiff, 54 pp.
- *Weber, W. J. - Diseases transmitted by rats and mice. Thompson Publications, P.O. Box 9335, Fresno, CA 93791, USA. 182 pp.
- Weber, W. - Rats and mice transmit more than 200 zoonotic pathogenic organisms. Proc. Indiana Acad. Sci. 92: 168 [Abstract]

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